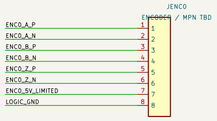
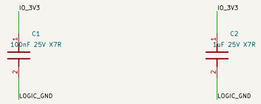
BASELINE TWO-ABZ DIFFERENTIAL RECEIVER — SCHEMATIC BLOCK r0.1

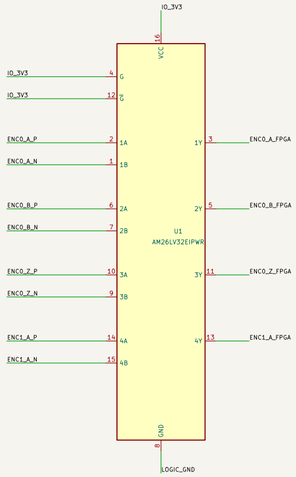
Six receiver channels, exact ledger allocation; no encoder supply limiter, line protection, FPGA pin assignment or board release.

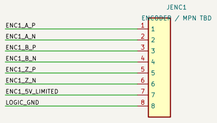
U1 = LEDGER RX0

AM26LV32EIPWR / PW16 / 3.0-3.6V supply
G=/G=3.3V: always enabled; receiver state does not grant motion.



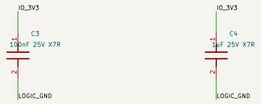


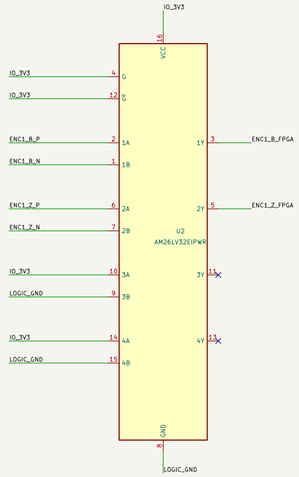




U2 = LEDGER RX1

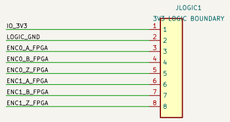
AM26LV32EIPWR / PW16 / 3.0-3.6V supply
G=/G=3.3V: always enabled; receiver state does not grant motion.

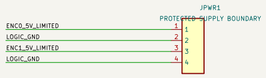




Unused U2 channels 3/4:
A tied IO_3V3; B tied LOGIC_GND.
Outputs 11/13 deliberately NC.
No floating or shorted unused inputs.

JENC pin1/2=A+/−; 3/4=B+/−; 5/6=Z+/−; 7=protected5V; 8=logic ground.
Supply limiters are absent; JPWR1 passes separately supplied, already protected rails.
JLOGIC1/JPWR1 are circuit boundaries only; connector MPNs remain unselected.





TERMINATION: fitted 120R per pair; DNP = open. Never substitute a short.
Location is receiver end only; encoder/cable profile must qualify the load.

















Termination reference: CRCW1210120RFKEA (1210,0.5W at70C).
Declared applied differential bound:5.5V; worst allowance −0.259W/channel.
Initial localambient<=70C: PCB/film temperature and load still require measurement.
Receiver +/−7V common mode does not rate termination for14V differential.
ESD component ratings do not qualify the complete controller.

TI SLLS849E: Fig5-1 / Table8-1 / Section10. Place C1/C3 close to U1/U2 pins16/8.
Open-circuit fail-safe does NOT guarantee a valid state for a shorted or terminated disconnected cable.
No galvanic isolation. Independent cable-loss detection or measured encoder timing here.

Partial schematic; no physical qualification or manufacturing release

Sheet: /
File: rx_abz.kicad_sch

Title: Baseline two-ABZ differential receiver

Size: A2	Date: 2026-09-28	**Rev: r0.1**
KiCad E.D.A. 10.0.4		Id: 1/1